

Workshop

Webapp Development

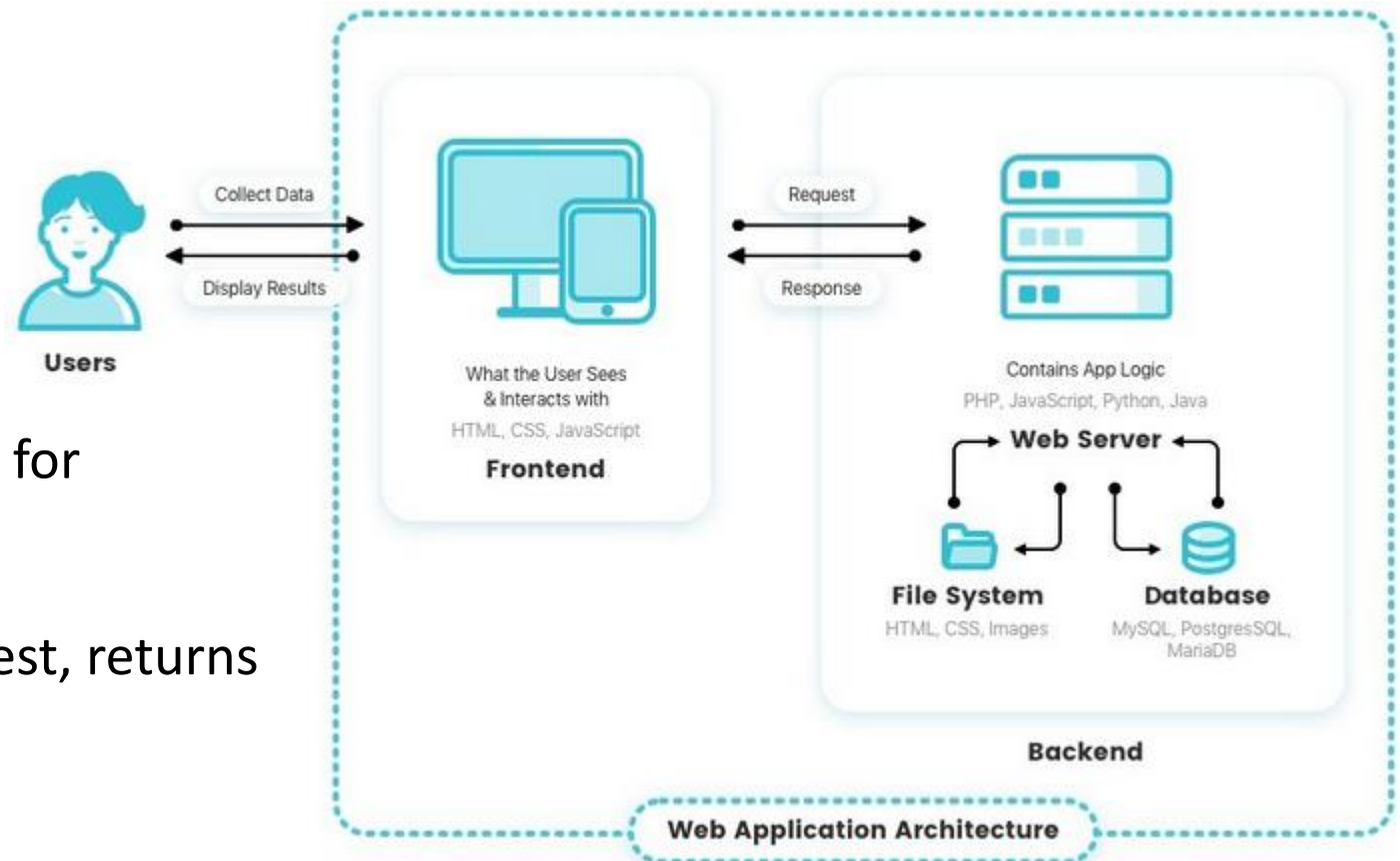
Using Spring Boot MVC

Today's Topic

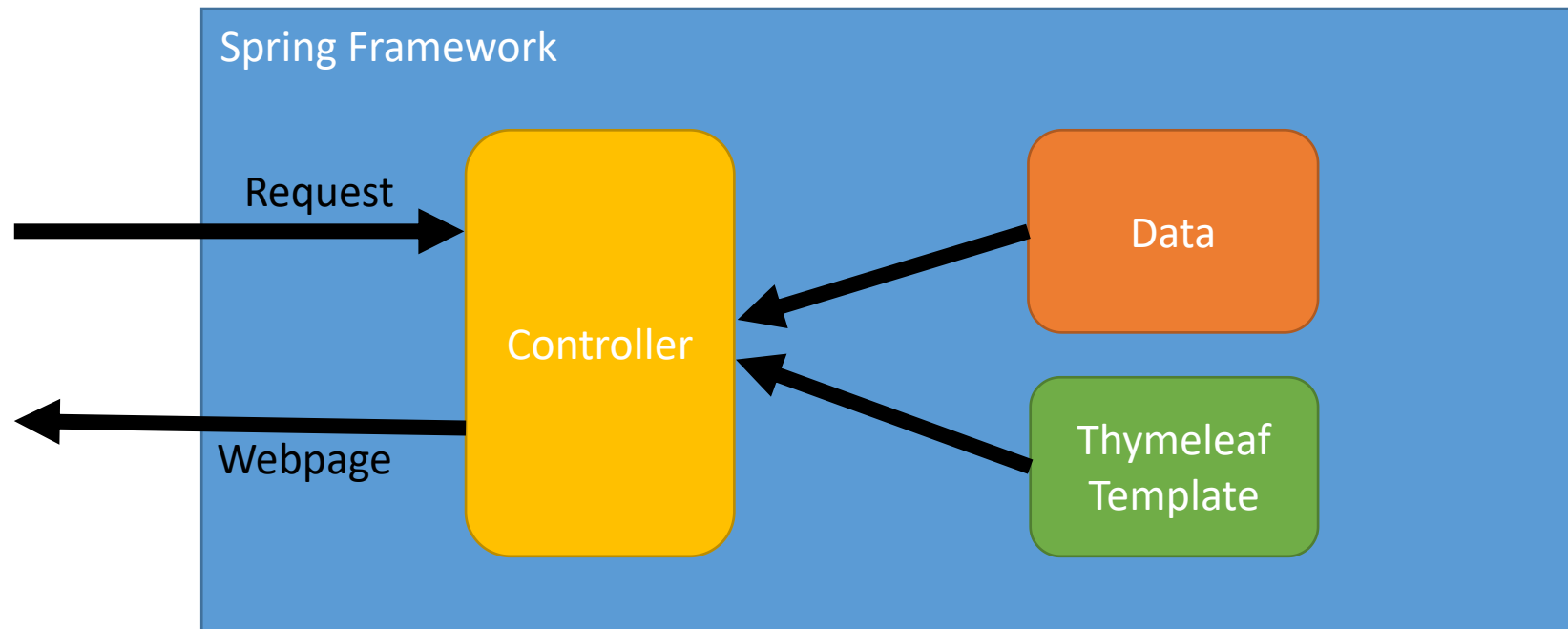
- Installing MS Visual Code and Spring Boot Extension Pack
- Installing MySQL Community Server and MySQLWorkbench
- Creating Web API using Spring Boot
- Accessing database from the web API

Webapp Architecture

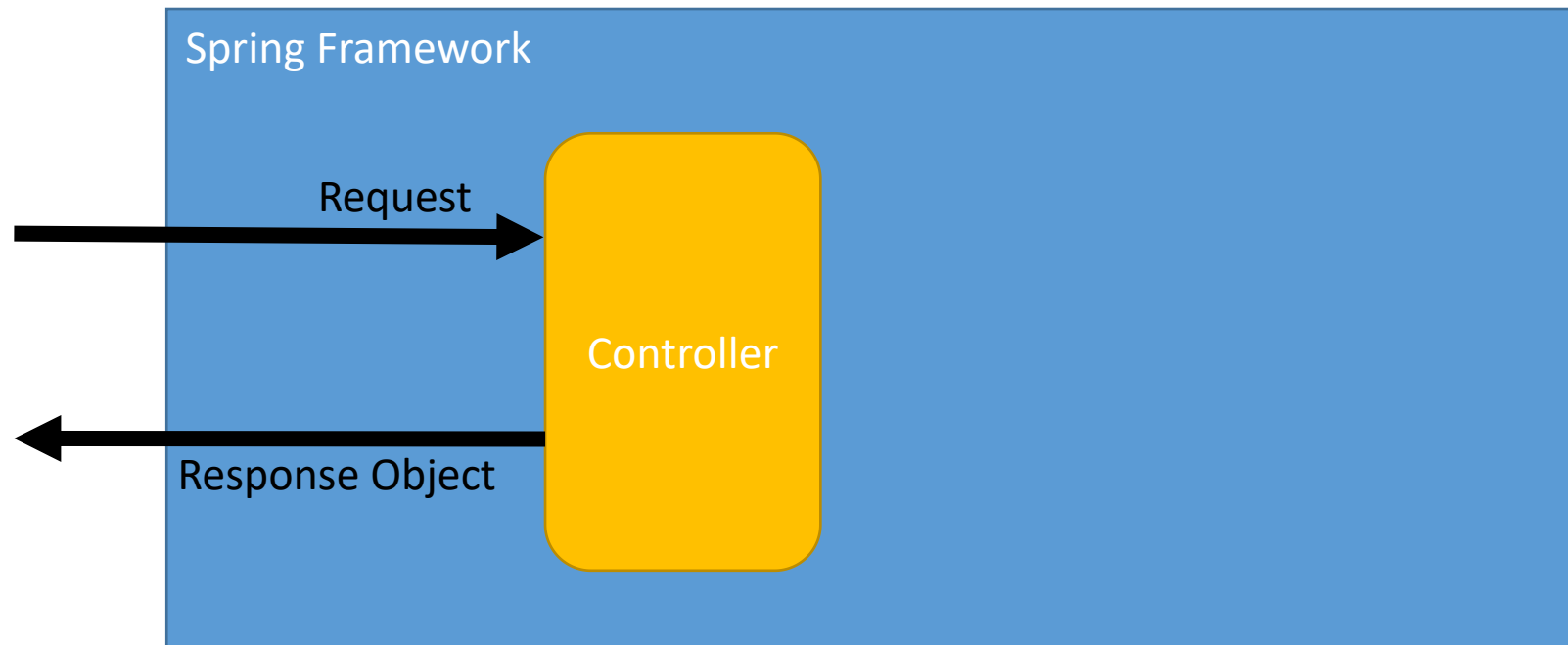
- Frontend
 - Browser
- HTTP Request
 - In the format of URL
 - POST, GET, etc
 - Frontend to Backend requesting for resources
- Backend
 - Web server: Accepts HTTP request, returns HTTP response
 - App server: Runs app logic
 - File server: Stores resources
 - Database server: Stores data



What is MVC?



Creating Web API using Spring Boot



Creating Web API using Spring Boot

- Before you can start the development, you need to prepare your development environment
 - Java sdk
 - My SQL Server & client
 - MS Visual Studio Code & Plugins

Creating Web API using Spring Boot

- Create a Spring Boot project using Spring Initializr.
- Add a RestController to the project to respond to a Request: -
 - GET
 - POST
 - PUT
 - DELETE
- Add RequestMapping, GetMapping, and PostMapping to map the endpoints to the java methods.
- Return a Response object to the request.
- Test the API using Postman. Observe the format of json using Postman.

Accessing database from the web API

- Reconfigure pom.xml
 - spring-boot-starter-data-jpa
 - mysql-connector-j
- Add a model to the project.
- Add a JPA repository interface to the project.
- Add a service object to access database using JPA repository interface.
- Access the service from the controller.
- Configure the database connection

Database Connection Configuration

```
spring.datasource.url = jdbc:mysql://localhost:3306/superhero
spring.datasource.username = root
spring.datasource.password = password
spring.datasource.driver-class-name= com.mysql.cj.jdbc.Driver
spring.jpa.database=mysql
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
spring.jpa.hibernate.naming-strategy=org.hibernate.cfg.ImprovedNamingStrategy
spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQL5Dialect
spring.session.store-type=jdbc
spring.session.jdbc.initialize-schema=always
```